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Crystallization Kinetics of Superionic Conductive Al(B, La)- Incorporated $LiTi_2(PO_4)_3$ Glass-Ceramics

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For the purpose of developing high-performance glass-ceramic superionic conductor, the controllable precipitation of $LiTi_2(PO_4)_3$-like superionic conducting phase in the Li_2O–TiO_2–P_2O_5 glass system was studied. Al with B or La co-incorporated $LiTi_2(PO_4)_3$-based glass-ceramics were prepared by the crystallization of the corresponding original glasses. Compared with the sole Al-incorporated $LiTi_2(PO_4)_3$-based glass-ceramics, the ionic conductivity shows an increase for the boron co-incorporated one and a decrease for the lanthanum co-incorporated one. Through the further in-depth analysis based on the methods of DSC and X-ray diffractive technique, this opposite change in ion conductivity was ascribed to the alterations of crystallization mechanism together with quantity of crystal phases within the glass-ceramics.. The boron addition promoted the precipitation of $LiTi_2(PO_4)_3$ phase and restrained the precipitation of second phase. The highest ionic conductivity 1.3×10^{-3} S/cm at 25°C was obtained through the heat treatment of B and Al co-incorporated glassy samples at 900° C for 12 h. These inorganic solid electrolytes have a potential application in lithium batteries or other electrochemical ionic devices.

I. Introduction

As a key material of all-solid-state ion devices, ionic conductive materials have been widely studied in the field of materials science, polymer science, electrochemistry, and solid-state chemistry. Considering their advantages of high safety assets, inflammability, easy fabrication, and wide electrochemical window stability compared with the liquid, colloidal, and polymer electrolytes, inorganic solid lithium ionic conductors have attracted much interest due to the potential applications in the field of high-energy density power sources and electrochemical ion devices.[1,2]

As a kind of important inorganic lithium-ion solid electrolytes, the NASICON-type Li ion conductors (systems derived from the $LiTi_2(PO_4)_3$ composition and partial substitution of Ti^{4+} by a trivalent cation, M^{3+}, leading to the $Li_{1+x}M_xTi_{2-x}(PO_4)_3$ system) has been extensively investigated.[3–5] Compared with the ceramic materials obtained by the classical sintering route, less porosity can be obtained through the glass-ceramic route, together with the easier and effective control of microstructure of the glass-ceramics, which can further increase the conductivity.[6–8] On the basis of these ideas, Narváez-Semanate and Rodrigues [9] investigated the crystallization process of the $16.2Li_2O$–$3.8Al_2O_3$–$42.5TiO_2$–$37.5P_2O_5$ glass, which shows a tendency for homogeneous nucleation and obtained a maximum ionic conductivity of 1.3×10^{-3} S/cm at room temperature through the glass-ceramic route. The introduction of Si^{4+} into $Li_{1+x}Al_xTi_{2-x}(PO_4)_3$ glass-ceramics leads to the formation of $2[Li_{1.8}Al_{0.4}Ti_{1.6}Si_{0.4}P_{2.6}O_{12}]\cdot AlPO_4$ glass-ceramics, whose maximum room-temperature conductivity as ascertained by Fu [10] can come up to 1.5×10^{-3} S/cm, however, the samples had obvious shortcoming that was easier to crack during heat treatment.[11] Through the further investigation of effects of M^{3+} ions on the conductivity of glasses and glass-ceramics in the system Li_2O–M_2O_3–GeO_2–P_2O_5 (M = Al, Ga, Y, Gd, Dy, and La),[12] Fu came to a conclusion that a great influence has taken place on the precipitation of main conductive crystalline phase of glass-ceramics by the incorporated trivalent ions and the formed unknown crystalline phases may hinder the migration of lithium ions, thus leading to a lower ionic conductivity. Accordingly, the controllable precipitation of $LiTi_2(PO_4)_3$-like superionic conducting phase is very important to develop high-performance glass-ceramic superionic conductors.

It is well known that as a good glass-former with low melting point, the trivalent oxide B_2O_3 can form the silicate-like network structure within the borophosphate glasses.[13] On the basis of these facts, in the present study, we investigated the effect of co-incorporation of Al_2O_3 with B_2O_3 or La_2O_3 into $LiTi_2(PO_4)_3$-based glasses upon the glass forming, crystallization behaviors and ionic conductivity. And we obtained a maximum room-temperature ionic conductivity 1.3×10^{-3} S/cm for the B_2O_3 and Al_2O_3 co-incorporated $LiTi_2(PO_4)_3$-based glass ceramic, which maybe originate from the effects of restraining the formation of the second phase and promoting the deposition of the main crystalline phase $LiTi_2(PO_4)_3$ due to the incorporation of B.

II. Experimental Procedure

(1) Glass Preparation and Crystallization Process

Three kinds of $LiTi_2(PO_4)_3$-based glass and glass-ceramics, with the molar percent of $14Li_2O$–$9Al_2O_3$–$38TiO_2$–$39P_2O_5$ (sample *a*), $14Li_2O$–$6Al_2O_3$–$3B_2O_3$–$38TiO_2$–$39P_2O_5$(sample *b*), and $14Li_2O$–$6Al_2O_3$–$3La_2O_3$–$38TiO_2$–$39P_2O_5$(sample *c*), were prepared in this study. Using the conventional melt quenching method, the original glassy samples were prepared through the commercial powder reagents of lithium carbonate Li_2CO_3(AR, $\geq 98.0\%$), aluminum hydroxide $Al(OH)_3$ (AR), boric acid H_3BO_3(AR, $\geq 99.5\%$), lanthanum oxide La_2O_3(SP, $\geq 99.95\%$), titanium oxide TiO_2(CP, $\geq 98.0\%$), and ammonium dihydrogen phosphate $NH_4H_2PO_4$(GR, $\geq 99.5\%$). After being sufficiently mixed according to the appropriate proportions, a 70 g batch was melted using a corundum crucible in an electrically heated furnace. Initially, the furnace was heated to 700°C and held for 2 h to release the volatile batch components, then raised to 1450°C and melted for 2 h. The melt obtained was rapidly poured into a steel plate preheated to 300°C, then pressed into a glass plate using another steel plate with thickness ~2 mm within a few seconds. Then the sample was annealed at 550°C for 6 h to relieve internal thermal stress, cooled naturally to room

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temperature within the furnace, through power shut off, and then a transparent purple glass plate was obtained.

Then the annealed bulk glass plates were cut and polished to get glassy samples with mirror surfaces on both sides. Subsequently, heat treatments were conducted at 700°C, 800°C, 900°C, 1000°C, and 1100°C for 12 h to obtain the corresponding glass-ceramic samples.

(2) Characterization of the Materials

The glass compositions were analyzed by ICP-AES (Optima 4300DV, PerkinElmer, Waltham, MA). Compared with the nominal composition, the analyzed content of Li decreases less significantly (about 5% from 14% to ~9%) due to volatilization during glass-forming process. Considering the corrosion effect of phosphate glass on corundum crucible, small amounts of Al_2O_3 enter into the glass, resulting in a slight increase in Al content, which is similar to the reference.[9] A little decrease in the B element occurs. Finally, the content of La_2O_3, TiO_2, and P_2O_5 changes slightly discrepancy due to experimental errors to chemical analysis. For the sake of the clarity, hereafter, the sample was also expressed by their nominal composition.

The glass transition (T_g), onset crystallization (T_i), and peak crystallization (T_p) temperatures of the glassy samples were characterized using differential scanning calorimetry (DSC, STA449c/3/G, Netzsch, Selb, Bavaria, Germany) within the temperature range 100°C–1000°C at different heating rates. The crystalline phases of the glass-ceramic samples were analyzed by X-ray diffraction (XRD) measurement using Cu Kα radiation (λ = 0.15418 nm, D/MAX-RB, Rigaku Corporation, Tokyo, Japan). For the electrochemical measurements, ~1 μm thick silver film was coated on both sides of these specimens with silver slurry and subsequently dried at 200°C for 2 h. The conductivity of the glass-ceramic sheets was measured by electrochemical a.c. impedance analyzer (CHI 760d, Shanghai Chenhua, Shanghai, China) with a small perturbation voltage of 5 mV in a frequency range from 10^6 to 1 Hz.

III. Results and Discussion

(1) Thermal Properties

Figure 1 shows DSC thermograms of the as-cast glass samples at a heating rate of 10°C/min. The first weak endothermic peak from 613.0°C to 624.5°C corresponds to the glass transition temperatures (T_g) of the glassy samples. Subsequently, the strong exothermic peaks (T_p) from 669.3°C to 720.0°C correspond to the crystallization of the glassy samples *a*, *b*, and *c*, respectively. Compared with sample *a* and *b*, two overlapped crystallization peaks T_{p1} and T_{p2} of sample *c* come out, which directly indicate the appearance of second crystalline phases in a some quantum. The temperature T_i stands for the onset temperature of crystallization and the temperature, T_f, for the completed temperature of crystallization. The values of T_g, T_i, and ΔT ($T_i - T_g$) are listed in Table I. It is seen that all the values of ΔT are small, which means the glass-forming tendency is comparatively poor in these compositions. Therefore, the glassy samples required to be quenched rapidly, otherwise it will crystallize extensively. Comparing the glassy sample *a*, the value ΔT of samples *b* and *c*, increased with the incorporation of boron or lanthanum, which means that the glass-forming tendency was enhanced by the incorporation of La_2O_3, especially B_2O_3.

According to the previous study,[13] the formation of BO_4 tetrahedra is superior to that of AlO_4 tetrahedra when aluminum and boron ions coexist within phosphate glasses, considering the facts of 3 or 4-coordination of the former and 4 or 6-coordination of the latter. When adding a small amount of B_2O_3 into the phosphate glass ($B_2O_3/P_2O_5 < 1$), B exists in the form of tetrahedral within the glassy network; and the BO_4 structural units are connected through the P–O bonds, thus leading to the formation of [BPO_4] groups. The existence of structure group [BPO_4] within the phosphate glass leads to stronger and denser glassy network structure, which will lower the expansion coefficient and enhance the thermal stability. This is the reason of the enhancement of glass-forming ability and thermal stability following the incorporation of boron in this study.

(2) Crystallization Behavior of the Prepared Glasses

Figure 2 shows DSC thermograms of the glass sample c recorded at different heating rates. With the increase in the heating rate, the glass transition temperature, and the peak crystallization temperature shift to higher temperatures. According to Ozawa,[14,15] the crystallization behavior of glasses can be investigated by the thermal analysis using different heating rates. The activation energy of crystalliza-

Table I. The Values of Characteristic Temperature, Calculated Parameters of Activation Energy E_c and Avrami Parameter n

	T_g (°C)	T_i (°C)	T_p (°C)	T_f (°C)	ΔT (°C) $T_i - T_g$	E_c (KJ/mol) Eq. 1	E_c (KJ/mol) Eq. 2	n
(a)	624.5	654.4	677.8	706.5	29.9	332	321	2.04
(b)	615.4	661.0	678.0	708.6	45.6	267	256	2.78
(c)	613.0	655.1	669.3[1]		42.1	277	265	2.35
			720.0[2]	731.4		504	492	1.84

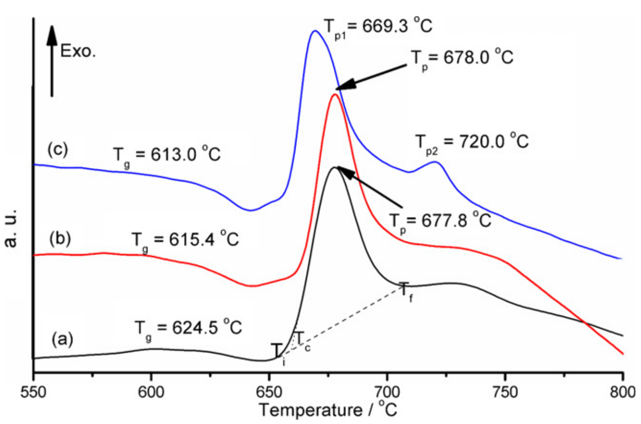

Fig. 1. Differential Scanning Calorimetry (DSC) thermograms of glassy samples with compositions (a) (b) and (c) measured at a rate of 10°C/min.

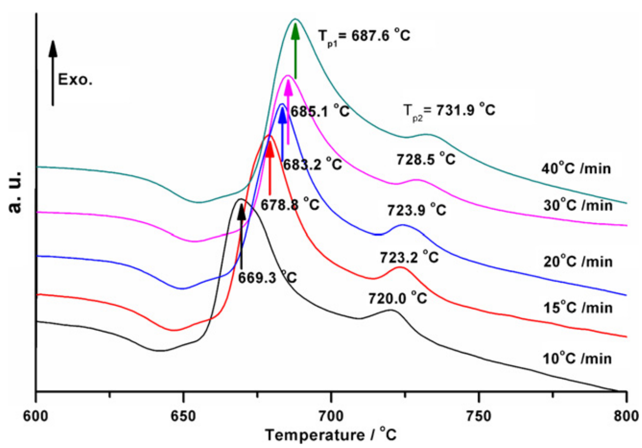

Fig. 2. DSC thermograms recorded at different heating rates for the glassy sample c.

tion, E_c, can be calculated using Ozawa plot via the following equation,

$$\ln(\phi) = -\frac{E_c}{RT_p} + \text{const} \tag{1}$$

where ϕ is the heating rate in the DSC measurement, E_c is the activation energy of crystallization, and T_p is the absolute crystallization peak temperature on DSC curves. According to the slopes of the straight lines shown in Fig. 3, the activation energy E_c of the crystallization process can be attained via the Eq. (1) and its values are listed in Table I. Here 2 values of E_c for the glassy sample *c* are originated from two crystallization peaks of this glass.

The activation energy of crystallization E_c can also be deduced from the variation in the peak crystallization temperature T_p with heating rate φ using the method of nonisothermal analysis derived by Kissenger *et al.*[16,17]:

$$\ln(\phi/T_p{}^2) = -E_c/RT_p + \text{const} \tag{2}$$

The plots of $\ln(\phi/T_p{}^2)$ vs. $1000/T_p$ are shown in Fig. 4. From the linear dependence, the activation energy of crystallization E_c for the three studied compositions is obtained to be 321, 256, and 265 (492) kJ/mol, respectively, and listed in Table I. Correspondingly, two activation energies of crystallization appear due to two crystallization peaks of the lanthanum-incorporated sample and the value of activation energy of crystallization of first crystalline phases is lower than that of the second crystalline phases.

It is well known in the glassy field, the glassy composition with higher glass-forming tendency shows a lower activation energy of crystallization in the ionic glassy system where the network former participates in the crystallization,[18] however a higher activation energy of crystallization in the corresponding covalent glassy system.[19] The present glassy system can be characterized as an overall ionic glassy system due to the existence of large amount of elements of higher ionicity. Surely, compared with the sample a, the incorporation of La^{3+} ions leads to a lower activation energy of crystallization and a better glass-forming tendency. Considering the effect of La^{3+} ions as an effective nucleating agent, the crystallization of the sample *c* will become easier, but meanwhile it leads to the precipitation of the second unexpected crystallization phase. When it comes to sample *b*, the incorporation of B element results in the formation of the 3-D network with rigid structural group [BPO_4] and a better glass-forming tendency. A lower activation energy of crystallization for this sample means that the addition of B into this glassy system is helpful to the migration of large ionic groups including B, P, and Ti ions to form a crystalline phase with NASICON-analogy structure, although it also increases lightly the crystallization peak temperature. The reduction in activation energy of crystallization following the incorporation of Boron is consistent with the previous reports.[20,21]

Furthermore, for the nonisothermal crystallization process, the volume fraction of crystallites *x*. heating rate φ, the Avrami exponent *n,* and the activation energy of crystallization E_a can be expressed using the following Matusita equation,[22]

$$\ln[-\ln(1-x)] = -n\ln(\phi) - 1.052mE_c/RT_p + \text{const} \tag{3}$$

where *m* and *n* are the constants related to the crystallization mechanism. The plots of $\ln[-\ln(1-x)]$ vs. $\ln\phi$ for these three glassy samples are also expected to be linear. According to the slope of the plots, the Avrami parameter (*n*) can be deduced. The experimental plots, fitting line, and parameters are shown in Fig. 5 and listed in Table I. To evaluate the mechanism of crystallization kinetics, the crystallization kinetics parameters *n* and *m* lying on the nucleation process and crystal growth, m can be deduced from $n = m + 1$.[23] According to the literature,[24] *n* may be 1, 2, 3 or 4, which are related to different crystallization mechanisms: $n = 1$ for surface nucleation and 1-D growth from surface to the inside; $n = 2$ for bulk nucleation, and 1-D growth; $n = 3$ for bulk nucleation and 2-D growth; $n = 4$ for bulk nucleation and 3-D growth.

In this study, for the glassy samples with the composition *a* and *c*, the values of *n* are approximately equal to 2, which means the crystallization mechanism of bulk nucleation and 1-D growth. When it comes to the glassy sample b, the parameter, n = 2.78 and close to 3, is higher than that of sample *a* and *c*, which means that the incorporation of B into the glass will change the crystallization process

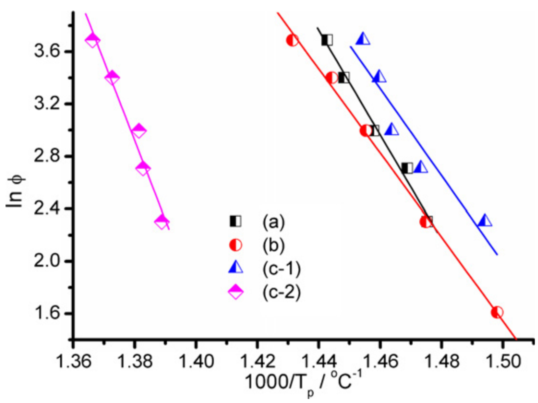

Fig. 3. ln(Φ) vs. 1000/T_p plots for the glassy samples (a) (b) and (c).

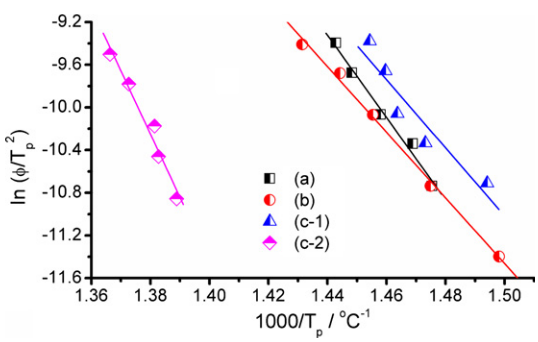

Fig. 4. ln($\Phi/T_p{}^2$) vs. 1000/T_p plots for the glassy samples (a) (b) and (c).

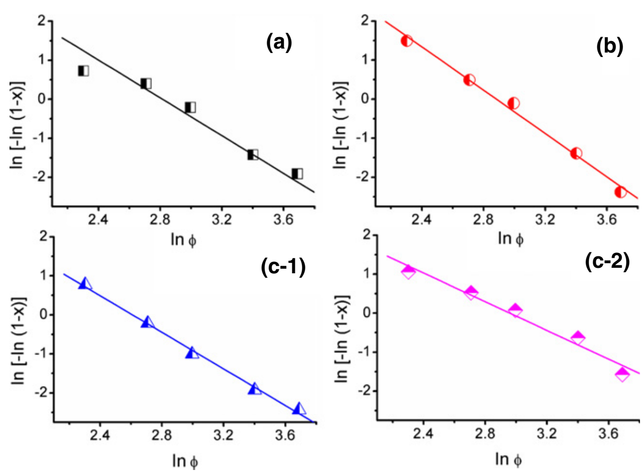

Fig. 5. Variation of ln[-ln(1−*x*)] with logarithm of heating rate for the glassy samples (a) (b) and (c).

from 1-D to 2-D crystal growth, maybe attributing to the descent of E_c. This is very conducive for the preparation of fast-ion conducting glass-ceramics. Nevertheless, further researches are needed for interpretation of this phenomenon.

(3) Crystalline Phases

X-ray diffraction patterns of the glass-ceramics with the compositions *a*, *b*, and *c* obtained by heat treatment at 900°C for 12 h are shown in Fig. 6. It is seen that the glass-ceramic samples had prominent diffraction peaks ascribed to the reflections of lithium-analogue of NASICON $LiTi_2(PO_4)_3$ phase[1,25,26] (JCPDS card 35-754) and other minor peaks consisting of $AlPO_4$, $LiTiPO_5$, $LaPO_4$, and TiO_2 phase. The subordinate crystalline phases of samples a and b obtained at 900°C for 12 h were $AlPO_4$, TiO_2, $LaPO_4$, and $LiTiPO_5$ phase appeared in the sample c for the same heat-treated process. In addition, enhancement in intensity of the diffraction peak for the secondary crystalline phase for sample c indicated the increase in content of impurity phases. Comparatively, least unexpected phase within sample *b* indicates the effects of restraining the formation of the second phase and promoting the deposition of the main crystalline phase $LiTi_2(PO_4)_3$ due to the incorporation of B, which is consistent with the discussion detailed in (2) section. As shown in Fig. 7, the as-cast glass exhibited only a wide smooth halo, which was a fingerprint of fully amorphous glassy phase; yet all the glass-ceramic samples that had prominent diffraction peaks belonged to lithium-analogue of NASICON $LiTi_2(PO_4)_3$ phase and other subordinate crystalline phases changed with different heat-treatment processes. The intensity of some strong peaks attributed to the presence of unexpected $AlPO_4$ and TiO_2 phases increased rapidly on the patterns of the glass-ceramics obtained at 1000°C and 1100°C as shown in Fig. 7.

(4) Electrochemical Properties

Using the sandwich configuration of Ag/sample/Ag, the ac impedance spectroscopy of samples is shown in Fig. 8 for the glass-ceramics obtained by annealing at 900°C or 1000°C of the three compositions. According to the previous report,[27] the impedance spectroscopy of the ceramic could be separated into two semicircles which attribute to different contributions, namely the bulk part, the grain-boundary part, and an inclined line corresponding to Warburg diffusion. Similar to the literatures[9,28] reported, only one semicircle was seen normally in the present glass-ceramics, another was very small and needed to be measured at lower temperatures (less than −10°C) and higher frequencies (>1 MHz). Meanwhile, the depressed semicircle was observed in the a.c. impedance

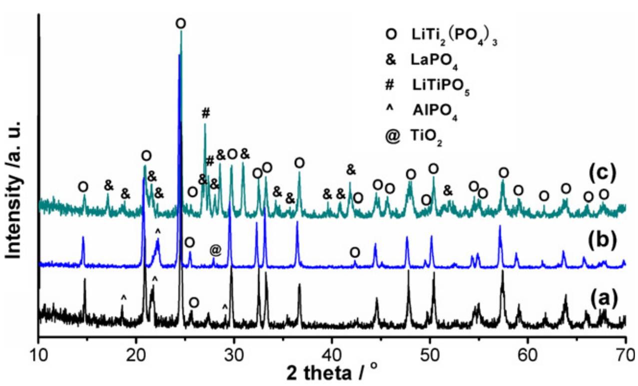

Fig. 6. XRD patterns of the glass-ceramics heat-treated at 900°C for 12 h for the compositions (a) (b) and (c).

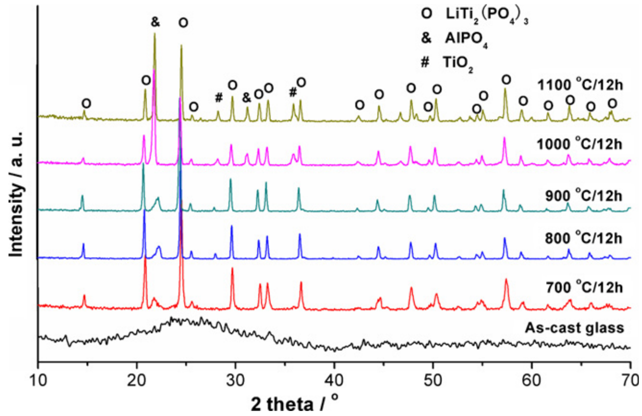

Fig. 7. XRD patterns of the glassy sample *b* and the derived glass-ceramics through heat-treatment at 700°C, 800°C, 900°C, 1000°C, and 1100°C for 12 h, respectively.

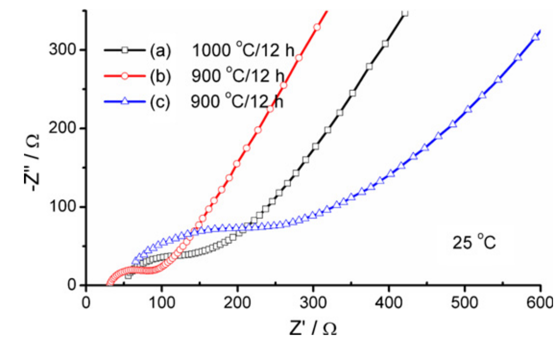

Fig. 8. Complex impedance plots of glass ceramics heat-treated at 900°C or 1000°C for 12 h for the compositions (a) (b) and (c).

spectroscopy of all samples, and this phenomenon could be explained by the "Jump relaxation model".[29]

According to the impedance value (R) determined by the impedance diagrams, the ionic conductivity can be calculated by the following formula:

$$\sigma = d/(R \times A)[\mathrm{S/cm}] \tag{4}$$

where d is the sample thickness, R is the electrolyte resistance, and A is the valid area. The calculated ionic conductivities of glass-ceramics obtained by annealing at different temperatures for 12 h are shown in Fig. 9. With the elevation of heat-treatment temperature from 700°C to 1100°C, the ionic conductivity first increased gradually, then after reaching the maximum began to decline slowly. For sample *a*, the maximum value can come up to 8.6×10^{-4} S/cm at 25°C when heat treated at 1000°C for 12 h, which is lower

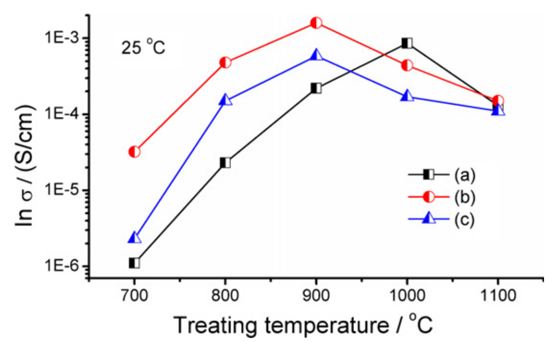

Fig. 9. Room-temperature ionic conductivity of glass-ceramics with compositions (a) (b) and (c) versus the heat-treatment temperature.

than the reported value of reference.[30] When it comes to sample c, we can get the maximum value of 5.85×10^{-4} S/cm. Finally, a maximum value of 1.3×10^{-3} S/cm can be obtained through heat treatment of sample b at 900°C for 12 h. Combined with the above discussion about the crystalline phase structures, the grains grew up gradually with the enhancement of crystallization temperature, so the grain boundary was reduced, thereby making more channels suitable for lithium ion migration, accordingly resulting in an increase in conductivity. When the heat-treatment temperature rises to 1000°C for sample a or 900°C for sample *b* and *c*, unexpected dielectric phases (TiO_2, $AlPO_4$, $LaPO_4$ and $LiTiPO_5$) with the blocking effect increased rapidly, so leading to the decrease in the ionic conductivity. Maybe just because of the least amount of impurity phases for the sample b, the boron-incorporation is beneficial to improve the ionic conductivity and a maximum value of 1.3×10^{-3} S/cm can be obtained through the appropriate crystallization process. As described above, sample *b* has the lowest active energy of crystallization and higher Avrami exponent, while crystallization behavior is dominated by the network former participated crystallization that is the precipitation of the lithium-analogue NASICON $LiTi_2(PO_4)_3$ phase. Therefore, the presence of boron can likely promote the crystallization of the main phase and reduce the crystalline temperature. On the other hand, the introduction of lanthanum into the glass may damage the equilibrium network structure[31] although it is also favorable to the network former participated crystallization. Thus, the evolution from a cross-linked ultraphosphate to chain-like meta-phosphate may lead to a phase separation, leading to the precipitation of more impurity phase and resulting in the decrease in conductivity. These results showed that the additives which promote the precipitation of the lithium-analogue NASICON phase and restrain the precipitation of second phase will increase the conductivity of the glass-ceramic.

IV. Conclusions

1. The Al(B, La)-incorporated $LiTi_2(PO_4)_3$ glass-ceramics were fabricated by crystallization of original glasses at 700°C–1100°C for 12 h, which contain the main crystalline phase of the lithium NASICON analogue $LiTi_2(PO_4)_3$ and minor phases of $AlPO_4$, TiO_2, $LaPO_4$, and $LiTiPO_5$. The optimum annealing temperature for making fast Li^+ ion-conductive glass-ceramic is found to be at 1000°C for only Al-incorporated composition and 900°C for the co-incorporated sample with B or La, thereafter $AlPO_4$, TiO_2, and other phases increased rapidly.
2. The co-introduction of B with Al results in the increase in glass-forming tendency and the decrease in activation energy of crystallization which crystallization mechanism can be 2-D growth with volume nucleation. Hence the co-introduction of B with Al promotes the precipitation of the main crystalline phase $LiTi_2(PO_4)_3$ and restrains the precipitation of the second phase, resulting in the higher room-temperature ionic conductivity of the glass-ceramic.
3. The co-introduction of La with Al also results in a little increase in glass-forming tendency and a little decrease in the activation energy for the crystallization of first crystalline phase. However, the precipitation of second phase was promoted and the crystallization is controlled by the mechanism of volume nucleation and 1-D growth, leading to a lower ionic conductivity of the corresponding glass-ceramics compared with sample *b*.

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